Quiz two

- Time: 9:30-11:50
- Toilet breaks are not permitted during the exam.
- Calculators are NOT allowed.
- Write your answers on the provided answer sheets. Additional sheets will be supplied upon request. Ensure that your full name is clearly written on every page of your answer sheets.
- Show all computation steps in sufficient detail. Marks are awarded for correct mathematical reasoning, intermediate calculations, and final answers. Missing or insufficient supporting calculations may result in deductions, even if the final answer is correct.

Question 1. (4 marks) Consider $\mathbb{R}^2$ together with addition and scalar multiplication defined as follows: for any $\boldsymbol{u} = \begin{pmatrix} u_1, & u_2 \end{pmatrix}$, $\boldsymbol{v} = \begin{pmatrix} v_1, & v_2 \end{pmatrix} \in \mathbb{R}^2$, and any $\alpha \in \mathbb{R}$

$$\boldsymbol{u} \oplus \boldsymbol{v} = \begin{pmatrix} u_1 + v_1 - 1, & u_2 + v_2 - 1 \end{pmatrix}, \quad \alpha\boldsymbol{u} = \begin{pmatrix} \alpha u_1, & \alpha u_2 \end{pmatrix}$$

(a) (0.5 marks) Compute $\boldsymbol{u} \oplus \boldsymbol{v}$ and $\alpha\boldsymbol{u}$ for $\boldsymbol{u} = \begin{pmatrix} 1, & 2 \end{pmatrix}$, $\boldsymbol{v} = \begin{pmatrix} 3, & 4 \end{pmatrix}$ and $\alpha = 6$.

(b) (0.5 marks) Show that $\begin{pmatrix} 0, & 0 \end{pmatrix}$ does not serve as the additive identity in $(\mathbb{R}^2, \oplus, \cdot)$.

(c) (1 mark) Prove that the additive identity in $(\mathbb{R}^2, \oplus, \cdot)$ is given by $\begin{pmatrix} 1, & 1 \end{pmatrix}$

(d) (1 mark) Find the additive inverse of any given $\boldsymbol{u} \in \mathbb{R}^2$

(e) (1 mark) Identify one vector space axiom that is not satisfied in $(\mathbb{R}^2, \oplus, \cdot)$. Clearly state the violated axiom and provide an example showing that it fails.

Solution.

(a) (0.5 marks)

$$\boldsymbol{u} \oplus \boldsymbol{v} = \begin{pmatrix} 1+3-1, & 2+4-1 \end{pmatrix} = \begin{pmatrix} 3, & 5 \end{pmatrix}, \quad \alpha\boldsymbol{u} = \begin{pmatrix} 6 \times 1, & 6 \times 2 \end{pmatrix} = \begin{pmatrix} 6, & 12 \end{pmatrix}$$

(b) Let $\boldsymbol{u} = \begin{pmatrix} 1, & 1 \end{pmatrix}$, then

$$\begin{pmatrix} 0, & 0 \end{pmatrix} \oplus \boldsymbol{u} = \begin{pmatrix} 0, & 0 \end{pmatrix} \neq \boldsymbol{u}. \quad \text{(0.5 marks)}$$

(c) For any $\boldsymbol{u} = \begin{pmatrix} u_1, & u_2 \end{pmatrix} \in \mathbb{R}^2$,

$$\boldsymbol{u} \oplus \begin{pmatrix} 1, & 1 \end{pmatrix} = \begin{pmatrix} u_1 + 1 - 1, & u_2 + 1 - 1 \end{pmatrix} = \boldsymbol{u} \text{(0.5 marks)}$$

similarly

$$\begin{pmatrix} 1, & 1 \end{pmatrix} \oplus \boldsymbol{u} = \boldsymbol{u} \quad \text{(0.5 marks)}$$

thus $\begin{pmatrix} 1, & 1 \end{pmatrix}$ is the additive identity in $(\mathbb{R}^2, \oplus, \cdot)$.

(d) For any $\boldsymbol{u} = \begin{pmatrix} u_1, & u_2 \end{pmatrix} \in \mathbb{R}^2$, let $\boldsymbol{w} = \begin{pmatrix} -u_1 + 2, & -u_2 + 2 \end{pmatrix}$, (0.5 marks) then

$$\boldsymbol{u} \oplus \boldsymbol{w} = \begin{pmatrix} u_1 + (-u_1 + 2) - 1, & u_2 + (-u_2 + 2) - 1 \end{pmatrix} = \begin{pmatrix} 1, & 1 \end{pmatrix}. \quad \text{(0.5 marks)}$$

Thus, $\boldsymbol{w}$ is the additive inverse of $\boldsymbol{u}$ in $(\mathbb{R}^2, \oplus, \cdot)$.

(e) Axiom: $\forall \boldsymbol{v}, \boldsymbol{u} \in V$ and $\forall \alpha \in \mathbb{R}$: $\alpha(\boldsymbol{u} \oplus \boldsymbol{v}) = \alpha\boldsymbol{u} \oplus \alpha\boldsymbol{v}$.
Let $\boldsymbol{u} = (1, \ 0)$, $\boldsymbol{v} = (0, \ 1)$, $\alpha = 2$ then

$$\alpha(\boldsymbol{u} \oplus \boldsymbol{v}) = 2\,(0, \ 0) = (0, \ 0)\,, \quad \alpha\boldsymbol{u} \oplus \alpha\boldsymbol{v} = (2, \ 0) \oplus (0, \ 2) = (1, \ 1)$$

which shows that the axiom does not hold.

Axiom: $\forall \boldsymbol{u} \in V$ and $\forall \alpha, \beta \in \mathbb{R}$: $(\alpha + \beta)\boldsymbol{u} = \alpha\boldsymbol{u} \oplus \beta\boldsymbol{u}$.
Let $\boldsymbol{u} = (1, \ 0)$, $\boldsymbol{v} = (0, \ 1)$, $\alpha = 2$, $\beta = 1$ then

$$(\alpha + \beta)\boldsymbol{u} = 3\,(1, \ 0) = (3, \ 0)\,, \quad \alpha\boldsymbol{u} \oplus \beta\boldsymbol{u} = (2, \ 0) \oplus (1, \ 0) = (2, \ -1)\,,$$

demonstrating that the axiom does not hold.
A correct identification of the failed axiom is worth (0.5 marks), and a proper example is worth (0.5 marks).

Question 2. (7 marks) Let $\mathcal{M}_{2\times 2}$ denote the set of all 2×2 matrices with coefficients from $\mathbb{R}$. For any $A = (a_{ij}), B = (b_{ij}) \in \mathcal{M}_{2\times 2}$ and any scalar $\alpha \in \mathbb{R}$, define matrix addition and scalar multiplication by

$$A + B = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{pmatrix}, \quad \alpha A = \begin{pmatrix} \alpha a_{11} & \alpha a_{12} \\ \alpha a_{21} & \alpha a_{22} \end{pmatrix}.$$

Together with those two operations, $\mathcal{M}_{2\times 2}$ forms a vector space over $\mathbb{R}$.

(a) (2 marks) Determine whether the following matrices are linearly independent in $\mathcal{M}_{2\times 2}$

$$\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$$

(b) (2.5 marks) Let $S = \{\, A_1, \ A_2, \ A_3, \ A_4 \,\}$, where

$$A_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \quad A_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

show that S is a basis for $\mathcal{M}_{2\times 2}$.

(c) (1.5 marks) Let

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

Find the coordinate vector of A relative to the basis S.

(d) (1 mark) Let $B = \{\, B_1, \ B_2, \ B_3, \ B_4 \,\}$, where

$$B_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad B_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

It can be verified that B forms a basis for $\mathcal{M}_{2\times 2}$. Find the transition matrix from the basis S to the basis B.

Solution.

(a) Consider the equation

$$\alpha_1 \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} + \alpha_2 \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Expanding both sides and equating the corresponding matrix entries, we obtain the system of linear equations

$$\begin{aligned} \alpha_1 + \alpha_2 &= 0 \\ 2\alpha_2 + \alpha_3 &= 0 \\ \alpha_1 + 2\alpha_2 + 2\alpha_3 &= 0 \\ 2\alpha_1 + \alpha_2 + \alpha_3 &= 0 \quad \text{(0.5 marks)} \end{aligned}$$

The augmented matrix of this linear system is

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 2 & 2 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right)$$

Applying Gauss–Jordan elimination, we obtain the reduced row echelon form:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \text{(1 mark)}$$

(−0.5 marks) if the student correctly sets up a sufficient system of linear equations and obtains the trivial solution, but does not show the intermediate calculation steps (e.g., Gaussian elimination or substitution).

Equivalent methods, such as substitution or Gaussian elimination, are accepted, provided that sufficient intermediate calculations are shown.

The system has only the trivial solution. It follows that the matrices are linearly independent in $\mathcal{M}_{2\times 2}$. (0.5 marks)

(b) To show S is a basis for $\mathcal{M}_{2\times 2}$, we must show that S linearly independent and span $\mathcal{M}_{2\times 2}$. To prove linear independence we must show that the equation

$$\alpha_1 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} + \alpha_4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{(0.5 marks)}$$

has only the trivial solution. Expanding both sides and equating corresponding entries, we obtain the following system of linear equations:

$$\begin{aligned} \alpha_1 &= 0 \\ \alpha_1 + \alpha_2 &= 0 \\ \alpha_1 + \alpha_2 + \alpha_3 &= 0 \\ \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 &= 0 \quad \text{(0.5 marks)} \end{aligned}$$

The coefficient matrix of the linear system is

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

Computing the determinant, we get $\det(M) = 1 \neq 0$. Since the determinant is nonzero, the system has only the trivial solution, which confirms that the matrices are linearly independent. (0.5 marks)

To prove that the matrices span $\mathcal{M}_{2\times 2}$ we must show that every 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ can be expressed as

$$\beta_1 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \beta_2 \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} + \beta_3 \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} + \beta_4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \quad \text{(0.5 marks)}$$

This corresponds to a system of linear equations with coefficient matrix M. Since $\det(M) \neq 0$, the system always has a unique solution for any given values a, b, c, d, proving that the given matrices span $\mathcal{M}_{2\times 2}$. (0.5 marks) Since S is both linearly independent and spans $\mathcal{M}_{2\times 2}$, it forms a basis for $\mathcal{M}_{2\times 2}$.

Alternative proofs using direct coefficient calculations or the dimension theorem are accepted. For a dimension-based argument, (−0.5 marks) if the student states $\dim(M_{2\times 2}) = 4$ without providing sufficient justification.

(c) Let $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$, we get the following system of linear equations in unknowns $\beta_1,\ \beta_2,\ \beta_3,\ \beta_4$

$$\begin{aligned} \beta_1 &= 1 \\ \beta_1 + \beta_2 &= 0 \\ \beta_1 + \beta_2 + \beta_3 &= 1 \\ \beta_1 + \beta_2 + \beta_3 + \beta_4 &= 0 \end{aligned}$$

Solving the system gives

$$\beta_1 = 1, \quad \beta_2 = -1, \quad \beta_3 = 1, \quad \beta_4 = -1$$

Thus, A can be expressed as

$$A = A_1 - A_2 + A_3 - A_4 \quad \text{(1 mark)}$$

Therefore, the coordinate vector of A relative to S is

$$[A]_S = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \text{ (0.5 marks)}$$

(d) Since

$$A_1 = B_1 + B_2 + B_3 + B_4$$

we have

$$[A_1]_B = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Similarly we get

$$[A_2]_B = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad [A_3]_B = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad [A_4]_B = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$P_{S\to B} = \begin{pmatrix} [A_1]_B & [A_2]_B & [A_3]_B & [A_4]_B \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} \quad \text{(1 mark)}$$

Question 3. (14 marks) Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3x_1 + 2x_2 \\ x_1 + x_2 \\ 0 \end{pmatrix}$$

Let A denote the standard matrix for T.

(a) (1 mark) Determine the matrix A.

(b) (2 marks) Find a basis for the row space of A.

(c) (1 mark) Find a basis for the range of T.

(d) (1 mark) Find a basis for the kernel of T.

(e) (1 mark) Determine the rank and nullity of A.

(f) (6 marks) Find the characteristic equation, the eigenvalues, and the bases for the eigenspaces of the matrix A.

(g) (2 marks) Use Doolittle's method to compute an LU decomposition of A.

Solution.

(a) The standard matrix is obtained by computing the images of the standard basis vectors $\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3 \in \mathbb{R}^3$

$$T\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}, \quad T\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad T\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore, the standard matrix is

$$A = \begin{pmatrix} 3 & 2 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{(1 mark)}$$

(b) The reduced row echelon form of A is

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{(1 mark)}$$

Thus a basis for the row space of A is

$$\{ (1, \ 0, \ 0), (0, \ 1, \ 0) \}. \quad \text{(1 mark)}$$

(c) Since the range of T corresponds to the column space of A, a basis for the range of T can be obtained from the pivot columns of A. By observing R, we see that a basis for the column space of A is

$$\left\{ \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\} \quad \text{(1 mark)}$$

(d) From R, we can deduce that a vector form for the general solution of the homogeneous system $A\boldsymbol{x} = \mathbf{0}$ is

$$\boldsymbol{x} = t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}. \quad \text{(0.5 marks)}$$

Thus a basis for the kernel of T is

$$\left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}. \quad \text{(0.5 marks)}$$

(e) $\text{rank}\,(A) = 2$, (0.5 marks) $\text{nullity}\,(A) = 1$. (0.5 marks)

(f) The characteristic equation of A is given by

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 3 & -2 & 0 \\ -1 & \lambda - 1 & 0 \\ 0 & 0 & \lambda \end{vmatrix} = \lambda((\lambda-3)(\lambda-1)-2) = \lambda(\lambda-(2-\sqrt{3}))(\lambda-(2+\sqrt{3})) = 0 \quad \text{(1 mark)}$$

Solving for λ, we obtain the eigenvalues:

$$\lambda_1 = 2 + \sqrt{3}, \quad \lambda_2 = 2 - \sqrt{3}, \quad \lambda_3 = 0 \quad \text{(0.5 marks)}$$

To determine a basis for the eigenspace corresponding to $\lambda_1 = 2 + \sqrt{3}$, we solve the homogeneous system

$$((2 + \sqrt{3})I - A)\boldsymbol{x} = \mathbf{0},$$

where

$$(2 + \sqrt{3})I - A = \begin{pmatrix} \sqrt{3} - 1 & -2 & 0 \\ -1 & \sqrt{3} + 1 & 0 \\ 0 & 0 & 2 + \sqrt{3} \end{pmatrix}. \quad \text{(0.5 marks)}$$

We find the general solution:

$$\boldsymbol{x} = t \begin{pmatrix} 1 + \sqrt{3} \\ 1 \\ 0 \end{pmatrix}. \quad \text{(0.5 marks)}$$

Thus, a basis for the eigenspace corresponding to $\lambda_1 = 2 + \sqrt{3}$ is

$$\left\{ \begin{pmatrix} 1 + \sqrt{3} \\ 1 \\ 0 \end{pmatrix} \right\}. \quad \text{(0.5 marks)}$$

To determine a basis for the eigenspace corresponding to $\lambda_2 = 2 - \sqrt{3}$, we solve the homogeneous system

$$((2 - \sqrt{3})I - A)\boldsymbol{x} = \mathbf{0},$$

where

$$(2-\sqrt{3})I - A = \begin{pmatrix} -1-\sqrt{3} & -2 & 0 \\ -1 & 1-\sqrt{3} & 0 \\ 0 & 0 & 2-\sqrt{3} \end{pmatrix}. \quad \text{(0.5 marks)}$$

We find the general solution:

$$\boldsymbol{x} = t \begin{pmatrix} 1-\sqrt{3} \\ 1 \\ 0 \end{pmatrix}. \quad \text{(0.5 marks)}$$

Thus, a basis for the eigenspace corresponding to $\lambda_2 = 2-\sqrt{3}$ is

$$\left\{ \begin{pmatrix} 1-\sqrt{3} \\ 1 \\ 0 \end{pmatrix} \right\}. \quad \text{(0.5 marks)}$$

Similarly, to determine a basis for the eigenspace corresponding to $\lambda_3 = 0$, we solve the homogeneous system

$$\begin{pmatrix} -3 & -2 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \boldsymbol{x} = \mathbf{0}. \quad \text{(0.5 marks)}$$

We obtain the general solution:

$$\boldsymbol{x} = t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad \text{(0.5 marks)}$$

Thus, a basis for the eigenspace corresponding to $\lambda_3 = 0$ is

$$\left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}. \quad \text{(0.5 marks)}$$

For each eigenvalue, (−0.5 marks) if the eigenvector equation $(\lambda I - A)x = 0$ is not explicitly set up, and another (−0.5 marks) if the intermediate calculations leading to the general solution are omitted.

(g)

$$A = \begin{pmatrix} 3 & 2 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix} \quad \text{(0.5 marks)}$$

First row

$$u_{11} = 3, \quad u_{12} = 2, \quad u_{13} = 0 \quad \text{(0.5 marks)}$$

Second row

$$\begin{aligned} \ell_{21}u_{11} &= 1 \\ \ell_{21}u_{12} + u_{22} &= 1 \\ \ell_{21}u_{13} + u_{23} &= 0 \end{aligned} \implies \begin{aligned} \ell_{21} &= \frac{1}{3} \\ u_{22} &= 1 - \frac{2}{3} = \frac{1}{3} \\ u_{23} &= 0 \end{aligned} \quad \text{(0.5 marks)}$$

Third row

$$\begin{array}{rcl} \ell_{31}u_{11} & = & 0 \\ \ell_{31}u_{12}+\ell_{32}u_{22} & = & 0 \\ \ell_{31}u_{13}+\ell_{32}u_{23}+u_{33} & = & 0 \end{array} \Longrightarrow \begin{array}{rcl} \ell_{31} & = & 0 \\ \ell_{32} & = & 0 \\ u_{33} & = & 0 \end{array} \quad \text{(0.5 marks)}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 3 & 2 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence $A = LU$.

(−1 marks) if the correct matrices L and U are provided without any intermediate calculations demonstrating Doolittle's method. Partial deductions may apply if only some of the required calculations are shown.